Protection can leave withholding or an empty family

Known A = (1, 0), B = (0, 1); larger is higher on both criteria.

Objective weights w = (1, 0); both protection budgets $q_1 = q_2 = 1/2$.

Criterion 2 remains protected although its objective weight is zero.

Report A>B

N = 1; $(E_1, E_2) = (0, 1)$

Criterion 1: 0 ≤ 1/2 PASS

Criterion 2: 1 > 1/2 FAIL

Weighted risk = 0

Error fractions: (0, 1)

Report B>A

N = 1; $(E_1, E_2) = (1, 0)$

Criterion 1: 1 > 1/2 FAIL

Criterion 2: 0 ≤ 1/2 PASS

Weighted risk = 1

Error fractions: (1, 0)

Hold the pair

N = 0; $(E_1, E_2) = (0, 0)$

Criterion 1: 0 ≤ 0 PASS

Criterion 2: 0 ≤ 0 PASS

Weighted risk = τ

Error fractions: undefined because N = 0

At most 2 grades

Protected family: {hold}

Exactly 2 grades

Empty: no admissible report

Count rows are $E_k \le q_k N$, with M = 1. At N = 0, error fractions are undefined.
Feasibility under the supplied point mass does not establish fairness or repeated-sampling error control.